FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2015

CONCEPT RECAPITULATION TEST - III

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 234**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 – 12) contains 4 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

Section-A (13 to 18) contains 3 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-C (01 – 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. A spaceship is sent to investigate a planet of mass M and radius R. While hanging motionless in space at a distance 5R from the centre of the planet, the ship fires an instrument package of mass m (m < < mass of spaceship) with speed v_0 at an angle θ w.r.t. radial line between the centre of planet and spaceship. For what value of θ, the package will just graze the surface of the planet. Given $v_0 = \sqrt{\frac{GM}{5R}}$

   (A) 30° (B) 60°
   (C) 37° (D) none of these

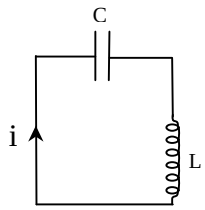
2. A charge distribution generates a radial electric field $\vec{E} = \frac{a}{r^2} e^{-\frac{r}{b}} \hat{r}$

   Where a and b are constants. The total charge giving rise to this electric field is
   (A) $4\pi\varepsilon_0 a$ (B) $4\pi\varepsilon_0 b$
   (C) $4\pi\varepsilon_0 ab$ (D) 0

3. Suppose an electrostatic potential has a maximum at point A and a minimum at point B. Choose the correct alternative:
   (A) point A is stable equilibrium point for both positive and negative charge
   (B) point B is stable equilibrium point for both positive and negative charge
   (C) point A is stable equilibrium point for positive and point B is stable equilibrium point for negative charge
   (D) point B is stable equilibrium point for positive and point A is stable equilibrium point for negative charge

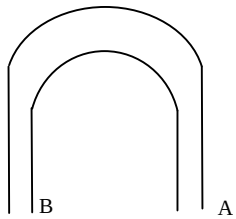
Space for rough work

4. Choose the correct alternatives for the circuit shown in figure.



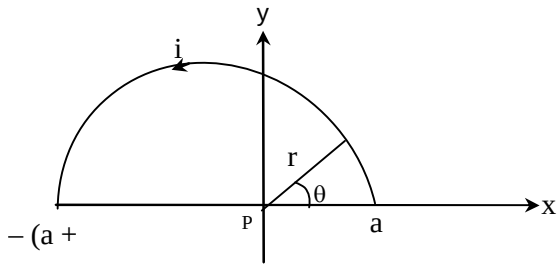
(A) The total energy is stored only in the electric field and is changing
(B) The total energy is stored only in the magnetic field and is changing
(C) The total energy is stored in both electric and magnetic field and is changing
(D) The total energy is stored in both electric and magnetic field and is constant

5. A glass rod of rectangular cross section is bent into the shape as shown in figure. The inner and outer contours of the curved portion of the rod have fixed radii R and R + a respectively. The refractive index of the glass is 1.5. For a given $R >> \lambda$, what is the maximum value of 'a' for which all the light entering at A will merge at B?



(A) R
(B) 0.5 R
(C) $\frac{R}{3}$
(D) 2 R

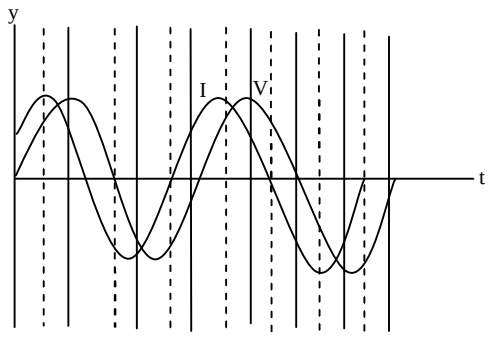
6. A wire segment is bent into the shape of an Archimedes spiral as shown in figure. The equation of the curve is, $r(\theta) = a + \frac{b}{\pi}\theta$ for $0 \le \theta < \pi$. The magnetic field at P is



(A) $\frac{\mu_0 i}{2r}\hat{k}$
(B) $\frac{\mu_0 i}{(2a+b)}\hat{k}$
(C) $\frac{\mu_0 i}{4b}\ln\left(1+\frac{b}{a}\right)\hat{k}$
(D) $\frac{\mu_0 i}{2b}\ln\left(1+\frac{b}{a}\right)\hat{k}$

Space for rough work

7. The graph shows current and voltage vs time in a driven RLC circuit at particular driving frequency. At this frequency the circuit is dominated by
(A) Inductance
(B) Capacitance
(C) Both (A) & (B)
(D) None of these



8. The volume of one mole of an ideal gas with adiabatic exponent γ is varied according to the law $V = \frac{a}{T}$, where a is a constant. Find the amount of heat obtained by the gas in this process if the gas temperature increased by ΔT.

(A) $\frac{2R\Delta T(2-\gamma)}{\gamma-1}$ (B) $\frac{R\Delta T(2-\gamma)}{\gamma-1}$

(C) $\frac{R\Delta T(2-\gamma)}{2(\gamma-1)}$ (D) $\frac{3R\Delta T(2-\gamma)}{2(\gamma-1)}$

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

9. A drum of mass m_1 and radius r_1 rotates freely with initial angular velocity ω_0. A second drum with mass m_2 and radius r_2 $(r_2 > r_1)$ is mounted on same axle and is at rest although it is free to rotate. A thin layer of sand with mass m is distributed on inner surface of smaller drum. At t = 0, small perforations in the inner drum are opened. The sand starts to fly out at a constant rate λ kg/sec and sticks to the outer drum. Ignore the transit time of the sand. Choose the correct alternatives.

(A) Angular speed of outer drum at time t is $\frac{\lambda t\omega_0}{m_2+\lambda t}\left(\frac{r_1}{r_2}\right)^2$

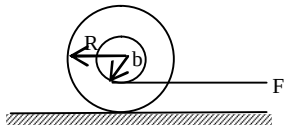
(B) Difference in final angular speeds of two drums is 0.

(C) Difference in final angular speeds of two drums is $\left(\frac{m(r_2^2-r_1^2)+m_2r_2^2}{(m+m_2)r_2^2}\right)\omega_0$

(D) Angular speed of inner drum remains constant.

Space for rough work

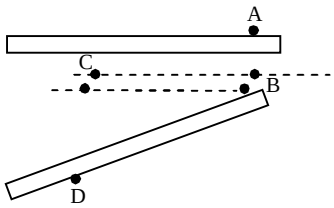
10. A Yo-Yo of mass m has an axle of radius b and a spool of radius R. The Yo-Yo is placed upright on a table and string is pulled with a horizontal force $\vec{F}$ to the right as shown in figure.



The static friction coefficient between Yo-Yo and table is μ_s. Now there are two cases.

case (a): The string is pulled very gently?
case (b): The string is jerked hard?
The Yo – Yo will rotate:
(A) Clockwise in case (a),
(B) Anticlockwise in case (a)
(C) Clockwise in case (b)
(D) Anticlockwise in case (b)

11. Holding the upper plate at + 5V relative to the ground of the lower plate, what is true about the electric potential at the following locations.



(A) V(A) > V(B)
(B) V(B) $\cong$ V(C)
(C) V(D) > V(A)
(D) V(C) > V(D)

12. The equation of a transverse wave travelling along a string is given by $y = 0.4\sin\left[\pi(0.5x - 200t)\right]$ where y and x are measured in cm and t in seconds. Suppose that you clamp the string at two points L cm apart and you observe a standing wave of the same wavelength as above transverse wave. For what values of L less than 10 cm is this possible?

(A) 2 cm (B) 4 cm
(C) 9 cm (D) 8 cm

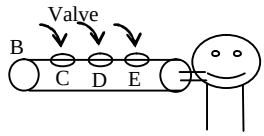
Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Questions 13 & 14

A flutes can be regarded as a tube open at both ends. It will emit a musical note if the flutist excites a standing wave in the air the column inside the tube. Speed of sound in air = 344 m/s.
$BC \cong CD \cong DE \cong 0.035$m.



13. The lowest musical note that can be played on flutes is C (261.7 H_2). What must be length of the tube? Assume that air column is vibrating in its fundamental mode.
(A) 0.251 m (B) 0.355 m
(C) 0.657 m (D) 0.87 m

14. In order to produce higher musical notes, the flutist, opens valves along the tube. Since the holes in these valves are large, an open valve has same effect as shorting tube. The flutist opens one valve to play C, two valves to play D, etc. find frequency of flutes when valve C is opened keeping D and E are closed. Assume flute vibrates in its fundamental mode.
(A) 276.5 hz (B) 305.5 hz
(C) 150.5 hz (D) 450.5 hz

Paragraph for Questions 15 & 16

A Rydberg hydrogenic atom is one in which the electron possess a very large quantum number e.g. n = 100. Take the electron charge to be – e (e > 0). The binding energy of Rydberg electron may be taken as $E_b = 10^{-3}$ eV, $e = 1.6 \times 10^{-19}$ C

15. What angular frequency of a photon can ionize such an atom?
(A) 10^{16} rad/s (B) 10^{10} rad/s
(C) 10^{8} rad/s (D) 10^{9} rad/s

16. Now consider the electron in the Rydberg hydrogenic atom to be unbounded and force for all pratical purpose. Supposing such Rydberg atoms are infected uniformly into an oscillatory electric field $F_0 \cos\omega t\, \hat{k}$. Provided by an electromagnetic wave, let the speed of the electron at time of infection (t = 0), be v = 0. Find maximum speed of electron.
(A) $\frac{eF_0}{2m\omega}$ (B) $\frac{eF_0}{3m\omega}$
(C) $\frac{eF_0}{5m\omega}$ (D) $\frac{eF_0}{m\omega}$

Space for rough work

Paragraph for Questions 17 & 18

When two slabs of N-type and P-type semiconductor are put in contact, the relative affinities of the materials cause electrons to migrate out of N-type material across the function to P-type material. This leaves behind a volume in the N-type material that is positively charged and orates a negatively charged volume in P-type material.

Let us model this as two infinite slabs of charge, both of thickness a with the function lying on the plane z = 0.

The N-type material lies in the range 0 < z < a and has uniform charge density $-\rho_0$. Thus:

$$\rho(x,y,z) = \rho(z) = \begin{cases} +\rho_0 & 0 < z < a \\ -\rho_0 & -a < z < 0 \\ 0 & |z| > a \end{cases}$$

The give answer of questions

17. Find electric field in religion –a < z < 0

(A) $\frac{\rho_0(a+z)}{\varepsilon_0}\hat{k}$ (B) $\frac{-2\rho_0(a+z)}{\varepsilon_0}\hat{k}$

(C) 0 (D) $\frac{-\rho_0(a-z)}{\varepsilon_0}\hat{k}$

18. Magnitude of potential difference between points z = –a to z = a

(A) $\frac{\rho_0 a^2}{4\varepsilon_0}$ (B) $\frac{+3\rho_0 a^2}{2\varepsilon_0}$

(C) $\frac{+\rho_0 a^2}{\varepsilon_0}$ (D) $\frac{\rho_0 a^2}{6\varepsilon_0}$

Space for rough work

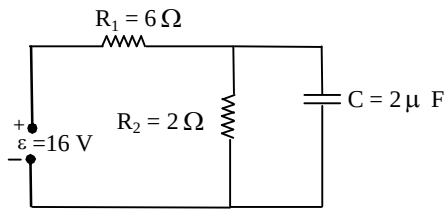
SECTION – C

(One Integer Value Correct Type)

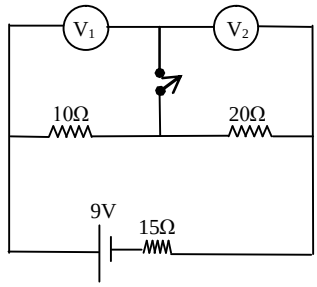
This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive)

1. A parallel plate capacitor of capacitance C is connected to a battery of emf ε until fully charged. It is then disconnected from battery and the plates are moved by a distance x so that potential difference between the plates is increased by four times. What is the volume of the dielectric (in m^3) necessary to fill the region between the plates.
Given C = 8.85 μF, x = 3 mm

2. Find maximum charge store on capacitor in μC.



3. In the circuit shown in the figure the emf of battery is 9v and its internal resistance is 15Ω. The two identical voltmeters can be considered ideal. Let V_1 and V_1^1 reading of 1st voltmeter when switch is open and closed respectively. Similarly, V_2 and V_2^1 be the reading of 2nd voltmeter when switch is open and closed respectively. Then find value of $\frac{V_2^1 - V_2}{V_1 - V_1^1} = ?$



4. A cube with sides of length $60\left(\sqrt{2}+1\right)$ cm and mass M is moving with an initial speed v_0 along a frictionless table. When it reaches the end of the table, it is caught abruptly by a short lip. What is the minimum speed v_0 in m/s such that it falls of the table?

5. Two beetles A and B hold the ends of a slightly stretched rubber band laid out on a horizontal table top. The initial positions of A and B are (0, 16 m) and (0, –4 m). A knot tied on the band is initially located at the origin. The beetle A starts running in +Y direction at constant acceleration and B moves in +X direction at constant speed of $2\sqrt{5}$ m/s. If the knot passes through point (8, 4), find the acceleration of A.

Space for rough work

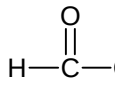
Chemistry

PART – II

SECTION – A

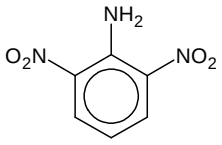
Straight Objective Type

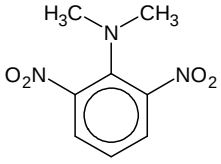
This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

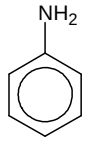
1.  , CH_3COOH can be separated by

(A) $NaHCO_3$
(B) Tollen's reagent
(C) Neutral $FeCl_3$
(D) All of these

2. Write the basicity order for following compounds

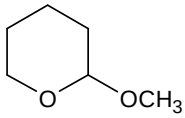
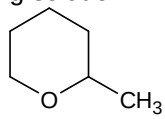
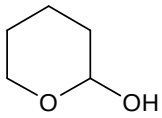
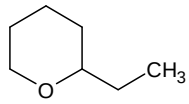

(I)


(II)


(III)

(A) III > II > I
(B) II > I > III
(C) II > III > I
(D) III > I > II

3. Which one of the following gives positive test with Fehling solution?

(A) 
(B) 
(C) 
(D) 

4. The radioactive decay $_{83}Bi^{211} \longrightarrow {}_{81}Tl^{207}$, takes place in 100 L closed vessel at 27°C. Starting with 2 moles of $_{83}Bi^{211}$ $\left(t_{1/2} = 130 \text{ sec}\right)$. The pressure development in the vessel after 520 sec will be

(A) 1.875 atm
(B) 0.2155 atm
(C) 0.4618 atm
(D) 4.618 atm

Space for rough work

5. The forward rate constant for the elementary reversible gaseous reaction

$C_2H_6 \rightleftharpoons 2CH_3$ is $1.57 \times 10^{-3} s^{-1}$ at 100 K

What is the rate constant for the backward reaction at this temperature if 10^{-4} moles of CH_3 and 10 moles of C_2H_6 are present in a 10 litre vessel at equilibrium

(A) 1.57×10^{6} L $mol^{-1}s^{-1}$ (B) 1.57×10^{10} L $mol^{-1}s^{-1}$

(C) 1.57×10^{11} L mol^{-1} s^{-1} (D) 1.57×10^{7} L mol^{-1} s^{-1}

6. A mixture of nitrogen and water vapours is admitted to a flask at 760 torr which contains sufficient solid drying agent after long time the pressure reached a steady value of 722 torr. If the experiment is done at 27°C and drying agent increase in weight by 0.9 gm, what is the volume of flask? Neglect any possible vapour pressure of drying agent and volume occupied by drying agent.

(A) 443.34 L (B) 246.3 L

(C) 12.315 L (D) 24.63 L

7. Among KO_2, AlO_2^-, BaO_2 and NO_2^+, unpaired electron is present in

(A) NO_2^+ and BaO_2 (B) KO_2 and AlO_2^-

(C) KO_2 only (D) BaO_2 only

8. Which of the following sentence is wrong about H_2O and HF

(A) Both the compounds resemble each other in solubility.

(B) Heat of formation of HF and H_2O are close to each other.

(C) Like H_2O, H_2F_2 is a weak acid and both are ionising solvents.

(D) The boiling points of HF and H_2O are high due to hydrogen bonding

Space for rough work

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct

9. Which of the following order is wrong with respect to alkali metals?
   (A) In gaseous state the order of reactivity is Li < Na < K < Rb < Cs.
   (B) conducting power $Cs^+ > Li^+ > K^+ > Na^+ > Rb^+$ (in aqueous Solution).
   (C) density order Li > Na > K > Rb > Cs.
   (D) In gaseous sate the order of reactivity is Li > Na > K > Rb > Cs.

10. Among the following statements, the correct statements are:
   (A) Calamine and cerrusite are carbonate ores
   (B) Rutile and cuprite are oxide ores
   (C) Zinc blende and pyrites are sulphides ores
   (D) malachite and azurite are sulphate ores of Cu

11. Which of the following is correct for an ideal solution?
   (A) $\Delta H_{mix} = 0$ and $\Delta V_{mix} = 0$ (B) $\Delta V_{mix} = 0$ and $\Delta S_{mix} > 0$
   (C) $\Delta H_{mix} > 0$ and $\Delta S_{mix} > 0$ (D) $\Delta G_{mix} = 0$ and $\Delta S_{mix} > 0$

12. Which of the following statements are correct
   (A) Benzene is more reactive than pyridine towards electrophilic aromatic substitution
   (B) Benzene is less reactive than pyrrole towards electrophilic aromatic substitution
   (C) Pyridine is more reactive than benzene towards nucleophilic aromatic substitution
   (D) Pyridine is more reactive than Benzene towards electrophilic aromatic substitution

Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Question Nos. 13 to 14

The cell potential (E_{cell}) of a reaction is related as $\Delta G = -nFE_{cell}$
For reversible reaction
at constant pressure $d(\Delta G) = -(\Delta S).dT$

$$\Delta G = \Delta H = T.\Delta S$$

$$\Delta G = \Delta H + T\left(\frac{d(\Delta G)}{dT}\right)_P$$

$\left[\frac{dE_{cell}}{dT}\right]_P$ is known as temperature coefficient of the e.m.f of the cell.

13. The temperature coefficient of the e.m.f of cell $\left(\frac{dE}{dT}\right)_P$ is given by

(A) $\frac{nF}{\Delta S}$ (B) $\frac{\Delta S}{nF}$

(C) $\frac{\Delta S}{nFT}$ (D) $-nFE$

14. At 300 K, ΔH for the reaction $Zn(s) + AgCl_{(s)} \longrightarrow ZnCl_2(aq) + 2Ag(s)$ is –218 kJ/mol. While the e.m.f. of the cell was 1.015 V. Then the value of $\left(\frac{dE}{dT}\right)_P$ for the cell is:

(A) -4.2×10^{-4} VK^{-1} (B) -3.81×10^{-4} VK^{-1}

(C) 0.11 VK^{-1} (D) 7.62×10^{-4} VK^{-1}

Space for rough work

Paragraph for Question Nos. 15 to 16

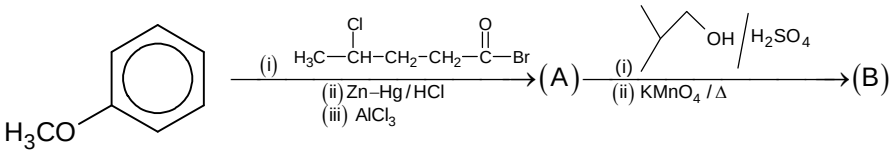
An unknown mixture contains one or two of the following: $CaCO_3$, $BaCl_2$, $AgNO_3$, Na_2SO_4, $ZnSO_4$ and NaOH. The mixture is completely soluble in water and solution gives pink colour with phenolphthalein. When dilute hydrochloride acid is gradually added to the solution, a precipitate is formed which dissolves with further addition of the acid.

15. Which of the following combination of compounds is soluble in water?
(A) $BaCl_2$ and $AgNO_3$ (B) $AgNO_3$ and NaOH
(C) $BaCl_2$ and Na_2SO_4 (D) $ZnSO_4$ and excess NaOH

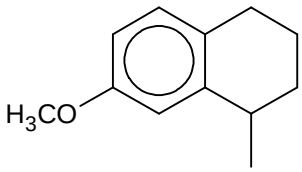
16. The aqueous solution of mixture gives white precipitate with dil HCl which dissolves in excess of dil. HCl. It confirms
(A) $BaCl_2$ + NaOH (B) Na_2SO_4 + NaOH
(C) $ZnSO_4$ + NaOH (D) $AgNO_3$ + NaOH

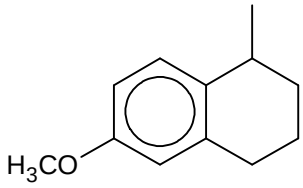
Space for rough work

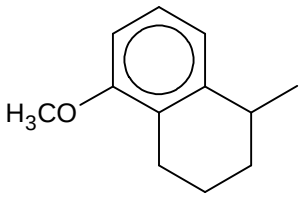
Paragraph for Question Nos. 17 to 18

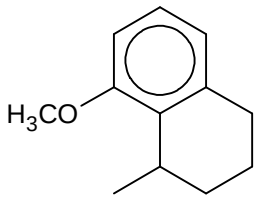


17. In the above sequence of reactions compound A is:

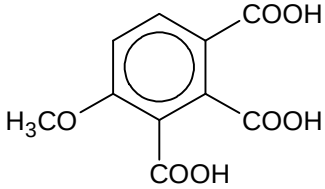
(A)


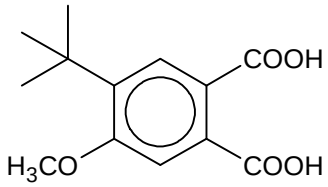
(B)


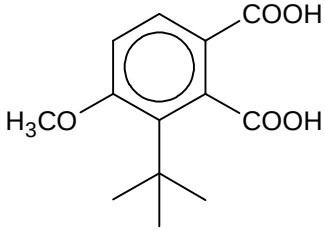
(C)


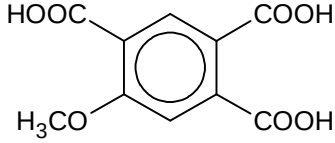
(D)


18. In the above sequence of reactions compound B is:

(A)


(B)


(C)


(D)


Space for rough work

SECTION – C

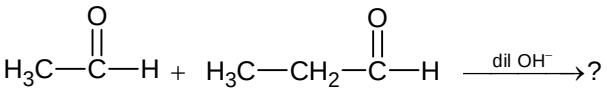
(One Integer Value Correct Type)

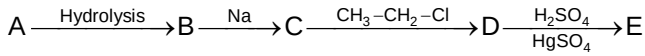
This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive)

1. If the lowest energy X-rays having $\lambda = 3.055 \times 10^{-8}$ m, estimate the minimum difference in energy between two Bohr's orbits such that an electronic transition would correspond to the emission of an X-ray. Assuming that the electrons in other shells exert no influence, at what Z (minimum) would transition from the second energy level to the first result in the emission of an X-ray?

2. A vessel of 250 litre was filled with 0.01 mole of Sb_2S_3 and 0.01 mole of H_2 to attain the equilibrium at 440°C as
$Sb_2S_3(s) + 3H_2(g) \rightleftharpoons 2Sb(s) + 3H_2S(g)$. In a separate experiment this H_2S is sufficient to precipitate 1.19 g PbS from Pb^{2+} solution. Then calculate K_c of the above reaction.

3. In metaborate structural arrangement, the number of B – O – B bonds is/are:

4. 
The number of possible products is/are:

5. 
E is 2-butanone then 'A' has how many π bonds?

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. If a_1 is the greatest value of f(x), where $f(x)=\left(\frac{1}{2+[\sin x]}\right)$ (where [.] denotes the greatest integer function) and $a_{n+1}=\frac{(-1)^{n+2}}{(n+1)}+a_n$, then $\lim_{n\to\infty} a_n$ is

(A) 1 (B) e^2
(C) ln 2 (D) none of these

2. If n be a positive integer such that $\sin\left(\frac{\pi}{2n}\right)+\cos\left(\frac{\pi}{2n}\right)=\frac{\sqrt{n}}{2}$, then

(A) $6 \le n \le 8$ (B) $4 < n \le 8$
(C) $4 \le n \le 8$ (D) $4 < n < 8$

3. Let f(x) = tan x and g(f(x)) = $f\left(x-\frac{\pi}{4}\right)$, where f(x) and g(x) are real values functions for all possible value of x, then f(g(x)) is

(A) $\tan\left(\frac{x-1}{x+1}\right)$ (B) tan (x – 1) – tan (x + 1)

(C) $\frac{f(x)+1}{f(x)-1}$ (D) $\frac{x-\frac{\pi}{4}}{x+\frac{\pi}{4}}$

Space for rough work

4. For which of the hyperbola, we can have more than one pair of perpendicular tangents?
(A) $\frac{x^2}{4}-\frac{y^2}{9}=1$ (B) $\frac{x^2}{4}-\frac{y^2}{9}=-1$
(C) $x^2 - y^2 = 4$ (D) $xy = 4$

5. A ray of light passing through the point P(1, 2) is reflected at a point Q on the x-axis and then passes through (5, 3), then the equation of reflected ray is
(A) $5x - 4y = 13$ (B) $5x - 4y = -3$
(C) $4x + 5y = 14$ (D) $4x - 5y = -6$

6. On the line segment joining (1, 0) and (3, 0) an equilateral triangle is drawn having its vertex in the fourth quadrant, then radical centre of the circles described on its sides as diameter is
(A) $\left(3, -\frac{1}{\sqrt{3}}\right)$ (B) $\left(3, -\sqrt{3}\right)$
(C) $\left(2, -\frac{1}{\sqrt{3}}\right)$ (D) $\left(2, -\sqrt{3}\right)$

7. Let P and Q be points (4, –4) and (9, 6) of the parabola $y^2 = 4a(x - b)$. Let R be a point on the arc of the parabola between P and Q, then the area of ΔPRQ is largest when
(A) $\angle PRQ = 90^o$ (B) the point R is (4, 4)
(C) the point R is $\left(\frac{1}{4}, 1\right)$ (D) the point R is $\left(1, \frac{1}{4}\right)$

8. If f(x) is a decreasing function then the set of values of 'k' for which the major axis of the ellipse $\frac{x^2}{f\left(k^2+2k+5\right)}+\frac{y^2}{f\left(k+11\right)}=1$ is the x-axis, is
(A) $k \in (-2, 3)$ (B) $k \in (-3, 2)$
(C) $k \in (-\infty, -3) \cup (2, \infty)$ (D) $k \in (-\infty, -2) \cup (3, \infty)$

Space for rough work

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct

9. If $\int \frac{\left(x^{-7/6}-x^{5/6}\right)}{x^{1/3}\left(x^2+x+1\right)^{1/2}-x^{1/2}\left(x^2+x+1\right)^{1/3}}dx=-P\left\{\frac{Z^3}{3}+\frac{3z^Q}{1}+\frac{3z^2}{T}+\log|z|\right\}+R$ where $z=\left(x+\frac{1}{x}+1\right)^{1/6}-1$, then

(A) P = 6
(B) Q – P = 4
(C) T + Q = 3
(D) Q – P + T = 5

10. If $\int e^{\tan^2 x}\sin 4x dx=-k\cos^4 x e^{\tan^2 x}+c$, then k is less than

(A) 3
(B) 4
(C) 5
(D) none of these

11. Let $f(t)=\int_{-\frac{\pi}{2}}^{t}\frac{\sin^3 x}{\sqrt[3]{\cos^2 x}}dx$, then

(A) $f\left(\frac{\pi}{2}\right)=0$
(B) $f(0)=-\frac{18}{7}$
(C) $f\left(\frac{\pi}{2}\right)=-\frac{18}{7}$
(D) none of these

12. If $f(x) = ax^2 + bx + c = 0$ has real roots and its coefficients are odd positive integers then
(A) f(x) = 0 always has irrational roots
(B) discriminant is a perfect square
(C) if ac = 1, then equation must have exactly one root 'α' such that $[\alpha] = -1$ (where [.] G.I.F)
(D) Equation f(x) = 0 has rational roots

Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Question Nos. 13 to 14

Read the following write up carefully and answer the following questions:

Let S be the set of points $\{(x, y): 0 \le x \le 3, 0 \le y \le 4, \text{where } x, y \in I\}$. Two distinct points are randomly chosen from S. It is given that mid-point of the line segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$ is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

13. The probability that the mid-point of the line segment of the two points chosen belongs to S, is

(A) $\frac{21}{95}$ (B) $\frac{42}{95}$

(C) $\frac{36}{95}$ (D) none of these

14. If two points chosen are (x_1, y_1) and (x_2, y_2), then the probability that $x_1 \ne x_2$ and $y_1 = y_2$ with $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) \in S$, is

(A) $\frac{8}{95}$ (B) $\frac{12}{95}$

(C) $\frac{14}{95}$ (D) none of these

Space for rough work

Paragraph for Question Nos. 15 to 16

Read the following write up carefully and answer the following questions:

Equation of the form P(sin x $\pm$ cos x, sin x cos x) = 0 where P(y, z) is a polynomial, can be valued by the change cos x $\pm$ sin x = t; 1 $\pm$ 2 sin x cos x = t^2. Reduce the given equation into $P\left(t, \frac{t^2-1}{2}\right) = 0$

15. If sin x + cos x = 1 + sin x cos x, then x is equal to

(A) $\frac{\pi}{2} + 2n\pi$ or $2n\pi$, $n \in I$ (B) $\frac{\pi}{4} + 2n\pi$ or $(2n + 1)\pi$, $n \in I$

(C) $2n\pi$, $n \in I$ (D) none of these

16. If $\sin^4 x + \cos^4 x = \sin x \cos x$, then x is equal to

(A) $(6n+1)\frac{\pi}{6}$ (B) $n\pi$

(C) $(4n+1)\frac{\pi}{4}$ (D) none of these

Paragraph for Question Nos. 17 to 18

Read the following write up carefully and answer the following questions:

Consider $f(x) = \frac{(x-\alpha)(x-\beta)}{(x-\gamma)(x-\delta)}$ where $0 < \alpha < \beta < \gamma < \delta$

17. Number of local maxima of f(x) will be

(A) 0 (B) 1

(C) 2 (D) can't be determined

18. Function f: D $\rightarrow$ R, (where D is the domain and R is the set of real numbers), will be

(A) one-one (B) many one

(C) bijective (D) none of these

Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive)

1. Let p(x) be a polynomial of degree 11 such that $p(x)=\frac{1}{1+x}$ for x = 0, 1, 2, ..., 11 then p(12) is _____

2. If the coefficient of the second, third and fourth terms in the expansion of $(1 + x)^n$ are in A.P. then n is _____

3. A tangent is drawn to the curve $y=\frac{8}{x^2}$ in xy-plane at the point $A(x_0, y_0)$ where $x_0 = 2$, and the tangent cuts the x-axis at a point B. Then $\overrightarrow{AB}\cdot\overrightarrow{OB}$ (where O is origin) is equal to _____

4. Let $f(x) = 30 – 2x – x^3$, then find the number of positive integral values of x which satisfies $f(f(f(x))) > f(f(–x))$ _____

5. If $x + \sin y = 2010$ and $x + 2010 \cos y = 2009$, $0 < y \le \frac{\pi}{2}$ then find the value of $\frac{[x+y]}{1005}$ (where [.] denotes the greatest integer function) _____

Space for rough work